

Title: Study of Frequency Division Multiplexing (FDM) using MATLAB

Objectives:

- To understand the basic concept and working principle of Frequency Division Multiplexing (FDM).
- To simulate multiple signals with different frequencies using MATLAB.
- To observe how different signals are combined into a single composite signal in FDM.
- To analyze the frequency spectrum of the multiplexed signal using MATLAB tools (e.g., FFT).
- To understand the process of signal separation (demultiplexing) from the combined signal.
- To study the effect of bandwidth allocation for multiple signals in FDM.
- To gain hands-on experience in MATLAB simulation of communication system concepts.

Theory:

Frequency Division Multiplexing (FDM) is a type of multiplexing where the available bandwidth of a communication channel is divided into multiple smaller frequency bands. Each signal is assigned a unique frequency range and transmitted simultaneously over the same channel. To prevent overlap and interference between adjacent signals, small unused spaces called *guard bands* are placed between frequency ranges. FDM is commonly used in analog communication systems such as radio broadcasting and television transmission, where multiple channels are sent at the same time without mixing.

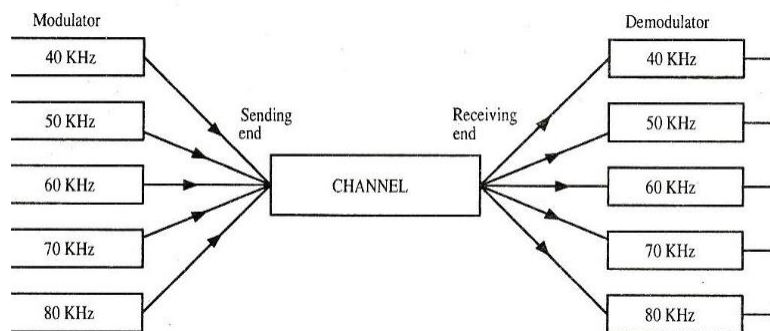


Fig: Frequency Division Multiplexing

Multiplexing is a communication technique in which multiple signals or data streams are combined and transmitted over a single shared communication channel. The main purpose of multiplexing is to efficiently utilize available bandwidth and reduce the cost of transmission. A device called a multiplexer (MUX) performs this process at the sender side by merging signals

from different sources into one composite signal. This combined signal travels through the channel without interfering with individual data streams because each is separated using a specific method such as frequency, time, or code.

Demultiplexing is the reverse process of multiplexing, performed at the receiver side. A device called a demultiplexer (DEMUX) receives the combined signal and separates it back into its original individual signals. It identifies each signal based on its assigned property (such as frequency in FDM) and directs it to the appropriate output channel. This ensures that each receiver gets only its intended data without interference from other signals.

Code:

```
clc;
clear;
close all;

fs = 4001;
t = 0:1/fs:1-1/fs;

% GIVEN VALUES (F=7, G=2)
am1 = 9; am2 = 12; am3 = 15; am4 = 18;
fm1 = 3; fm2 = 4; fm3 = 5; fm4 = 6;
Ac1 = 1; Ac2 = 1; Ac3 = 1; Ac4 = 1;

fc1 = 50;
fc2 = 100;
fc3 = 150;
fc4 = 200;

m1 = am1*cos(2*pi*fm1*t);
m2 = am2*cos(2*pi*fm2*t);
m3 = am3*cos(2*pi*fm3*t);
m4 = am4*cos(2*pi*fm4*t);

c1 = Ac1*cos(2*pi*fc1*t);
c2 = Ac2*cos(2*pi*fc2*t);
c3 = Ac3*cos(2*pi*fc3*t);
c4 = Ac4*cos(2*pi*fc4*t);

x = m1.*c1 + m2.*c2 + m3.*c3 + m4.*c4;
```

```

figure;
subplot(5,1,1); plot(t,m1); title('Message Signal 1');xlabel("Time"); ylabel("Amplitude");
subplot(5,1,2); plot(t,m2); title('Message Signal 2');xlabel("Time"); ylabel("Amplitude");
subplot(5,1,3); plot(t,m3); title('Message Signal 3');xlabel("Time"); ylabel("Amplitude");
subplot(5,1,4); plot(t,m4); title('Message Signal 4');xlabel("Time"); ylabel("Amplitude");
subplot(5,1,5); plot(t,x); title('Composite FDM Signal');xlabel("Time"); ylabel("Amplitude");

[a,b] = butter(5, [(fc1-fm1-6)/(fs/2),(fc1+fm1+6)/(fs/2)]);
bpf1 = filter(a,b,x);
[c,d] = butter(5, [(fc2-fm2-6)/(fs/2),(fc2+fm2+6)/(fs/2)]);
bpf2 = filter(c,d,x);
[e,f] = butter(5, [(fc3-fm3-6)/(fs/2),(fc3+fm3+6)/(fs/2)]);
bpf3 = filter(e,f,x);
[g,h] = butter(5, [(fc4-fm4-6)/(fs/2),(fc4+fm4+6)/(fs/2)]);
bpf4 = filter(g,h,x);

z1 = 2*bpf1.*c1;
z2 = 2*bpf2.*c2;
z3 = 2*bpf3.*c3;
z4 = 2*bpf4.*c4;

[i,j] = butter(5, (fm1+3)/(fs/2));
rec1 = filter(i,j,z1);
[k,l] = butter(5, (fm2+3)/(fs/2));
rec2 = filter(k,l,z2);
[m,n] = butter(5, (fm3+3)/(fs/2));
rec3 = filter(m,n,z3);
[o,p] = butter(5, (fm4+3)/(fs/2));
rec4 = filter(o,p,z4);

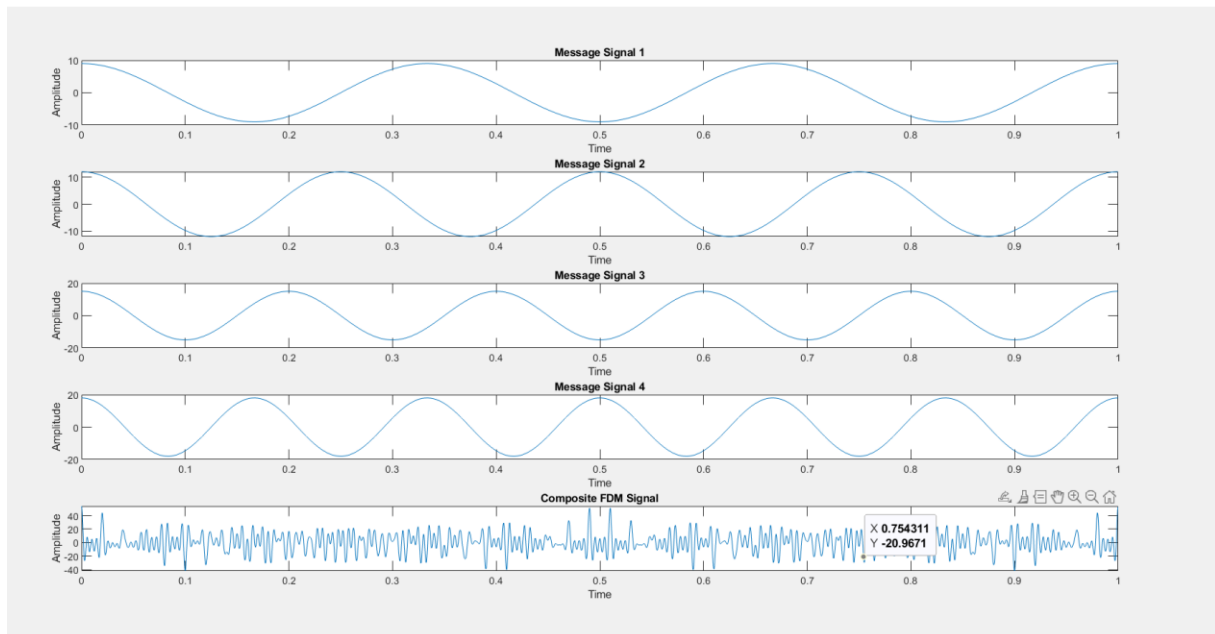
```

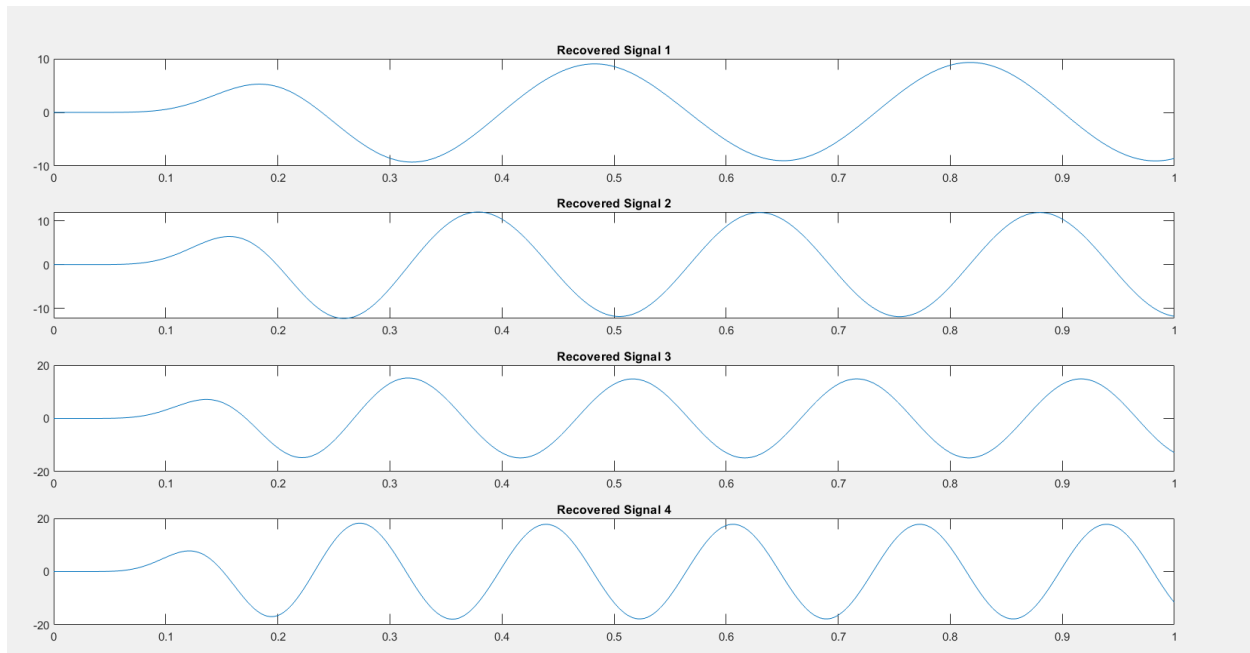
```

figure;
subplot(4,1,1); plot(t,rec1); title('Recovered Signal 1');
subplot(4,1,2); plot(t,rec2); title('Recovered Signal 2');
subplot(4,1,3); plot(t,rec3); title('Recovered Signal 3');
subplot(4,1,4); plot(t,rec4); title('Recovered Signal 4');

```

Simulation Figure:





Result analysis/discussion:

The MATLAB based experiment demonstrated that multiple message signals can be effectively transmitted together using Frequency Division Multiplexing. Each message has a unique carrier frequency when combining the composite spectrum displayed distinct peaks at the carrier frequencies, confirming theoretical principle. During the demultiplexing process band pass filters were used to extract each modulated Signal from the composite spectrum Proper filter design ensured minimal cross-channel interference and multiplying the filtered signals with their respective carriers shifted them back to baseband for recovery. The recovered signals in both time and frequency domains closely matched the original messages. The experiment also highlighted the significance. of bandwidth allocation and guard bands. When carrier frequencies are placed too close to each other, overlapping spectra can cause inter channel interference and distortion during signal separation. The reinforces the need for careful carrier frequency planning in real-world applications such as broad casting, satellite and mobile communication.

Conclusion: Maisha

References:

[1] MathWorks, *MATLAB Documentation*, Natick, MA, USA: MathWorks. [Online]. Available: <https://www.mathworks.com/help/>

[2] Stormy Attaway, *MATLAB: A Practical Introduction to Programming and Problem Solving*, 6th ed. Boston, MA, USA: Butterworth-Heinemann, 2019